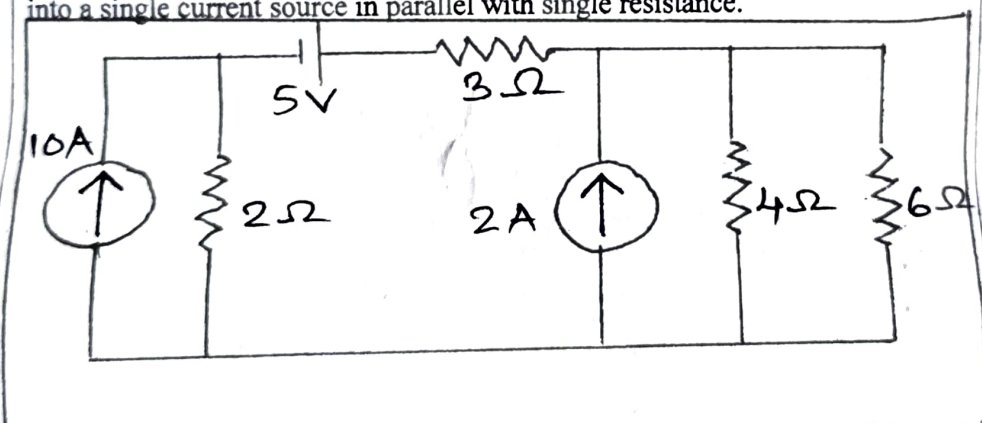
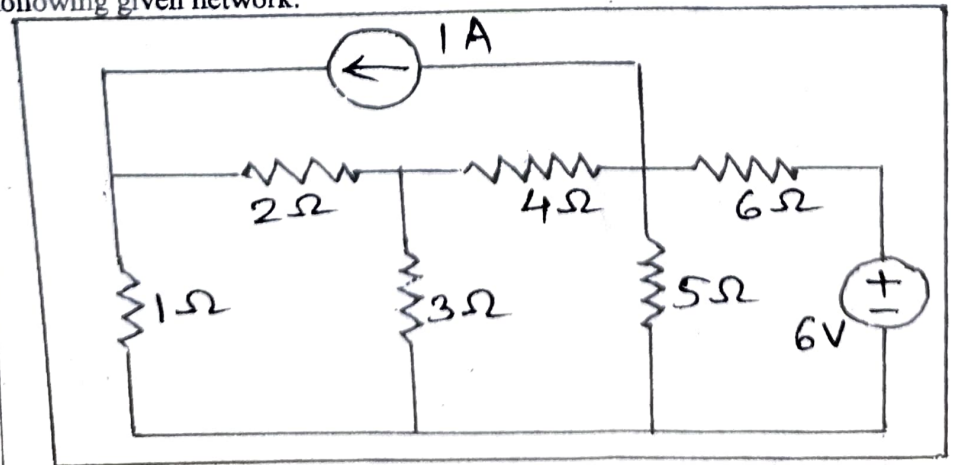
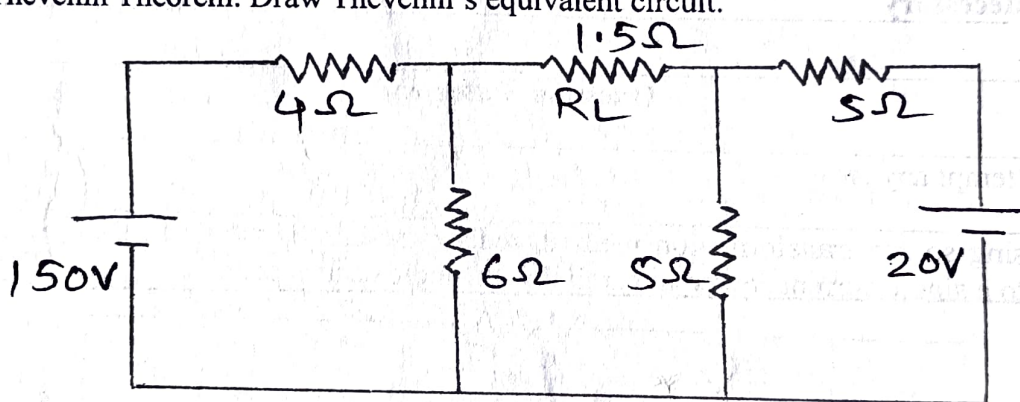




Semester: January 2025– April 2025 Supplementary Exam August 2025 Examination: End-Semester Examination			Duration: 2 Hrs.
Maximum Marks: 50	Programme code: 06 Programme: B.Tech	Class: FY	Semester: II(SVU 2023)
Institute/School/Department: K. J. Somaiya School of Engineering		Name of the department: ETRX/EXTC/IT/MECH/RAI/CCE/EXCP /VLSI	
Course Code: 216U06C106	Name of the Course: Elements of Electrical and Electronics Engineering		
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any two	
i)	Using source transformation method, reduce the circuit shown in the figure into a single current source in parallel with single resistance. 	05
ii)	Using Node analysis method determine the current through 2 Ω resistor in the following given network. 	05



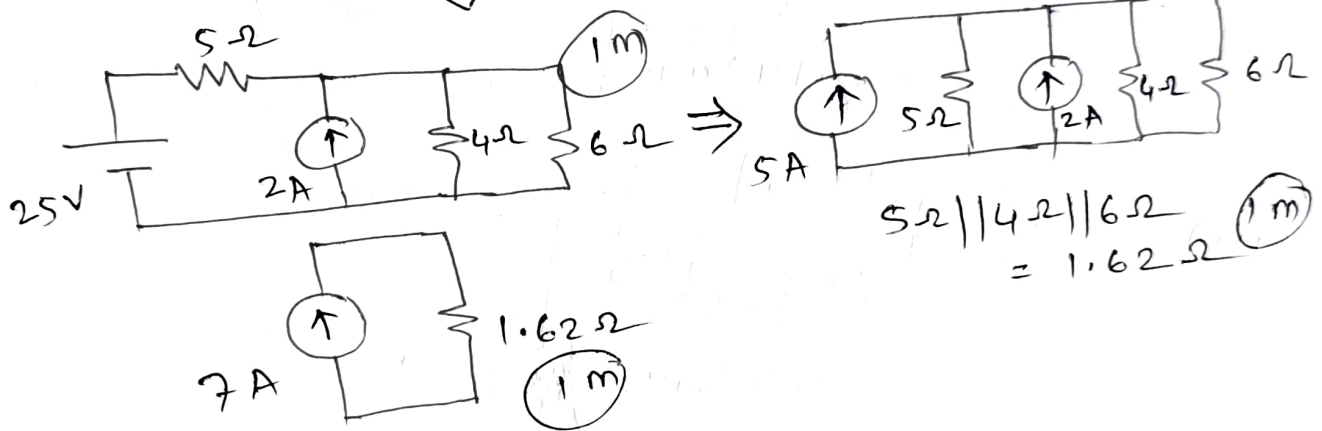
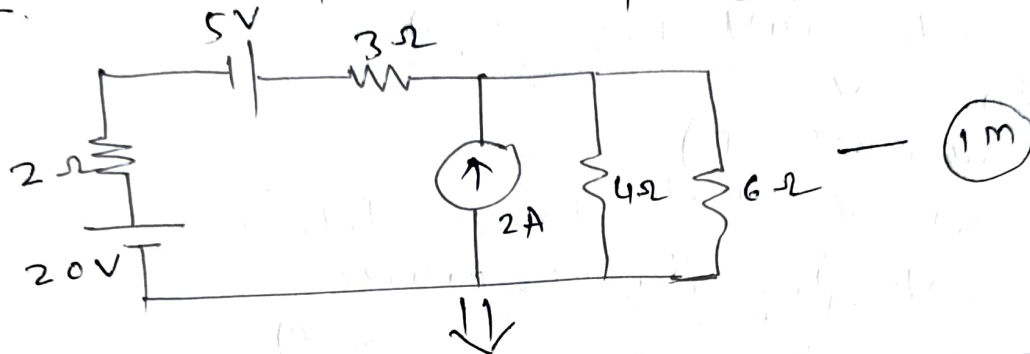
iii)	Two impedances $(14+j5) \Omega$ and $(18+j10) \Omega$ are connected in parallel across 200V, 50Hz supply. Determine: 1) Admittance of each branch 2) Total admittance of the circuit 3) Total current 4) Power factor of the circuit 5) Comment on the nature of the pf of the circuit	05
Q.2	Attempt any one	10
i)	Determine the current through resistance $R_L = 1.5 \Omega$ in the circuit by using Thevenin Theorem. Draw Thevenin's equivalent circuit. 	10
ii)	Each phase of a three phase delta connected load has impedance of $Z_{ph} = (50 \angle 60^\circ) \Omega$. The line voltage is 440V. Calculate the total power. What will be the readings of the two wattmeters connected to measure the power?	10
Q.3	Attempt any one	10
i)	A series R-L-C circuit has the following parameter values: $R = 10\Omega$, $L = 0.014H$, $C = 100\mu F$ Calculate: 1) Resonant frequency in rad/sec and in Hz 2) Quality factor of the circuit 3) Lower and upper cut off frequencies 4) Bandwidth 5) Maximum value of the voltage appearing across the capacitor if the voltage $v = 20\sin 2000t$ is applied across the R-L-C series circuit.	10
ii)	Three similar coils connected in star type load, take total power of 1.5kW with pf of 0.2 lagging from a three phase 440 V, 50Hz supply. Calculate resistance and inductance of each coil.	10
Q.4	Attempt the following	10



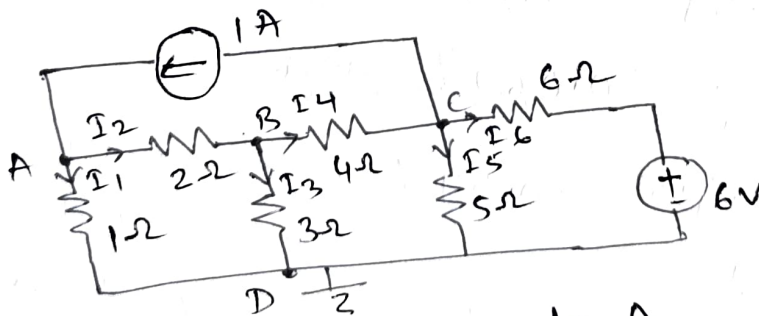
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i)	Draw circuit diagram for CE configuration and input-output characteristics of the circuit with correct labels.	05
ii)	Draw and explain in brief single phase induction motor.	05
Q.5	Attempt the following	10
i)	Justify the use of Zener diode as a voltage regulator.	05
ii)	Write the equations/values for the following parameters of Full Wave Bridge rectifier 1) DC load current 2) RMS load Voltage 3) Ripple Factor 4) Transformer Utilisation Factor 5) Rectifier Efficiency	05

Q.1 (i) $10A \parallel 2\Omega \rightarrow \text{equt. ckt-}$



(ii)



Applying KCL at node A

$$1 = I_1 + I_2$$

$$1 = \frac{V_A}{1} + \frac{V_A - V_B}{2} \quad (1m)$$

$$3V_A - V_B = 2 \quad (1)$$

Applying KCL at node B

$$I_2 = I_3 + I_4$$

$$\frac{V_A - V_B}{2} = \frac{V_B}{3} + \frac{V_B - V_C}{4} \quad (1m)$$

$$6V_A - 13V_B + 3V_C = 0 \quad (2)$$

Applying KCL at node c

$$I_4 = I_5 + I_6 + 1$$

$$\frac{V_B - V_C}{4} = \frac{V_C}{5} + \frac{V_C - 6}{6} + 1 \quad (1m)$$

$$30V_B - 74V_C = 0 \quad (3)$$

$$V_A = 0.803V$$

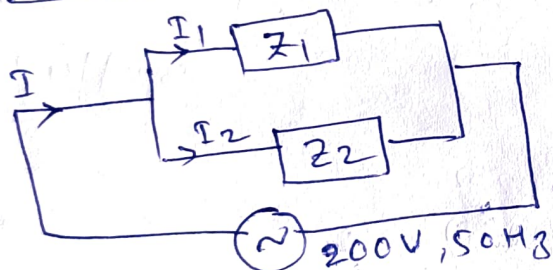
$$V_B = 0.4088V \quad (1m)$$

$$V_C = 0.165V$$

$$I_{2\Omega} = \frac{V_A - V_B}{2} = \frac{0.803 - 0.4088}{2} \quad (1m)$$

$$I_{2\Omega} = 0.19Amp$$

(iii)



$$Z_1 = 14 + 5j = 14.87 \angle 19.65^\circ \Omega$$

$$(1) \therefore Y_1 = \frac{1}{Z_1} = 0.0673 \angle -19.65^\circ S \quad (1m)$$

$$Z_2 = 18 + j10 = 20.59 \angle 29.05^\circ \Omega$$

$$Y_2 = \frac{1}{Z_2}$$

$$(1) (ii) Y_2 = 0.0486 \angle -29.05^\circ S \quad (1m)$$

$$(2) \text{ Total admittance } = Y_T = Y_1 + Y_2$$

$$Y_T = 0.115 \angle -23.59^\circ S \quad (1m)$$

$$(3) I = \frac{V}{Z_T} = V \cdot Y_T \quad (1m)$$

$$I = 23.1 \angle -23.59^\circ Amp$$

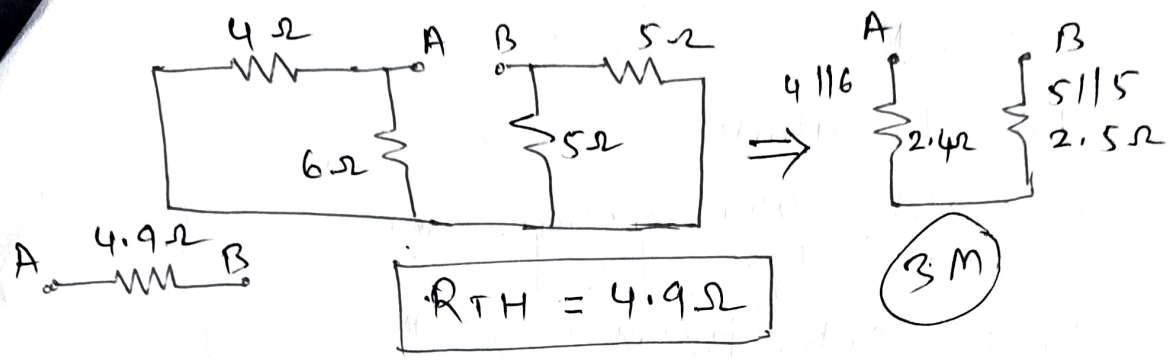
$$(4) P.F.1 = \cos(-19.65) = 0.942 \quad (1m)$$

$$P.F.2 = \cos(-29.05) = 0.874 \quad (1m)$$

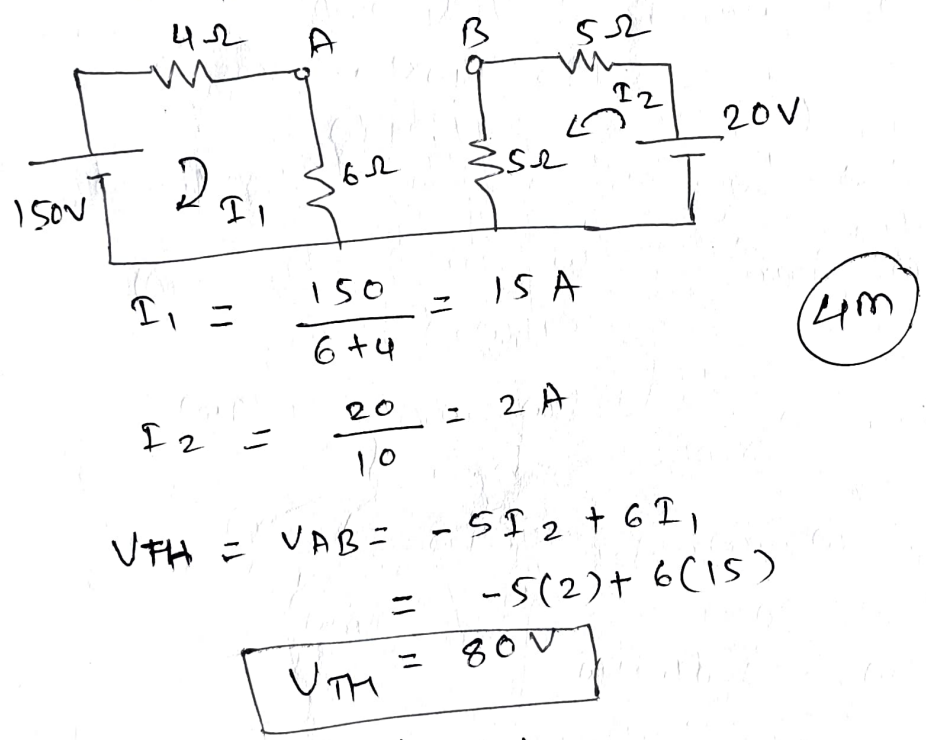
$$(5) \text{ lagging P.F.} \quad (1m)$$

$$\text{Tot. PF} = 0.916$$

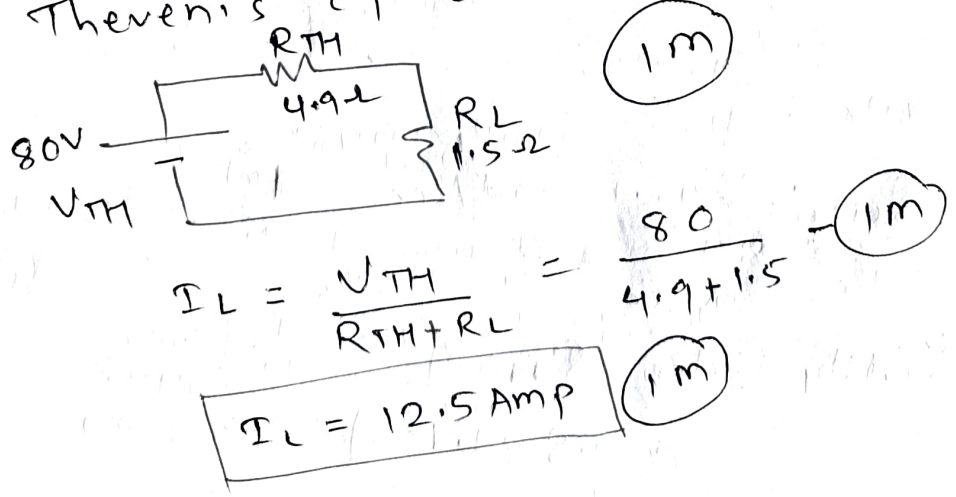
2 (i) Calculation of R_{TH}



Calculation of V_{TH}



Theremi's eq^t ckt



Q.2 (ii) $\bar{Z}_{ph} = (50 \angle 60^\circ) \Omega$, $V_L = 440V$, load - Delta

For A load

$$V_{ph} = V_L = 440V \quad - (1m)$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{440}{50} = 8.8 \text{ Amp} \quad - (1m)$$

$$I_L = \sqrt{3} I_{ph} = \sqrt{3} \times 8.8 \quad - (1m)$$

$$I_L = 15.24 \text{ Amp}$$

$$\begin{aligned} \text{Total Power} = P &= \sqrt{3} V_L I_L \cos \phi \\ &= \sqrt{3} \times 440 \times 15.24 \times \cos 60 \\ &= 5807.21 \text{ W} \end{aligned} \quad (2m)$$

$$\text{Total Power} = P = W_1 + W_2 = 5807.21 \text{ W} \quad (1m)$$

$$\phi = \tan^{-1} \left\{ \frac{\sqrt{3}(\omega_2 - \omega_1)}{(\omega_1 + \omega_2)} \right\} \quad - (1m)$$

$$\tan 60 = \left\{ \frac{\sqrt{3}(\omega_2 - \omega_1)}{5807.21} \right\} \quad - (1m)$$

$$\therefore \omega_2 - \omega_1 = 5807.21 \text{ W} \quad - (1m)$$

$$\therefore \begin{cases} W_1 = 0 \text{ W} \\ W_2 = 5807.21 \text{ W} \end{cases} \quad - (1m)$$

Q.3 (i) $\omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.014 \times 100 \times 10^{-6}}}$

$$\omega_r = 845.15 \text{ rad/sec} \quad - (1m)$$

$$f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi} \omega_r = 134.5 \text{ Hz}$$

$$f_r = 134.5 \text{ Hz}$$

$$- (1m)$$

$$\text{Quality factor} = Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{10} \sqrt{\frac{0.014}{100 \times 10^{-6}}}$$

$$Q = 1.183 \quad - (2m)$$

$$\text{Bandwidth} = B = \frac{R}{L} = \frac{10}{0.014} \quad - (1m)$$

$$B = 714.29 \text{ rad/sec}$$

lower cut-off freq. $\omega_1 = \omega_r - \frac{\beta\omega}{2}$

$$\boxed{\omega_1 = 488 \text{ rad/sec}} \quad (1m)$$

Upper cut-off freq $\omega_2 = \omega_r + \frac{\beta\omega}{2}$

$$\boxed{\omega_2 = 1202.3 \text{ rad/sec}} \quad (1m)$$

Applied volg = $v = 20 \sin 2000t$ volts

$$V = \frac{V_m}{\sqrt{2}} = \frac{20}{\sqrt{2}}$$

$$V = 14.14 \text{ V} \quad (1m)$$

Max. volg across cap^r = $V_c = \frac{V}{R} \sqrt{\frac{L}{C}}$

$$V_c = \frac{14.14}{10} \sqrt{\frac{0.014}{100 \times 10^{-6}}} \quad (2m)$$

$$\boxed{V_c = 16.73 \text{ V}} \quad (1m)$$

Q.3 (ii) $V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{440}{\sqrt{3}} \quad (1m)$

$$V_{ph} = 254.03 \text{ V}$$

$$P = \sqrt{3} V_L I_L \cos \phi \quad (1m)$$

$$1.5 \times 10^3 = \sqrt{3} \times 440 \times I_L \times 0.2 \quad (1m)$$

$$\therefore I_L = 9.84 \text{ A}$$

In star connection, $I_{ph} = I_L = 9.84 \text{ A} \quad (1m)$

$$Z_{ph} = \frac{V_{ph}}{I_{ph}} = 25.82 \Omega \quad (1m)$$

$$\text{phase angle } \phi = \cos^{-1} 0.2 \quad (1m)$$

$$\therefore \phi = 78.46^\circ$$

$$\overline{Z_{ph}} = (25.82 \angle 78.46^\circ) \Omega \quad (1m)$$

$$\overline{Z_{ph}} = (5.17 + j25.3) \Omega \quad (1m)$$

$$\therefore \boxed{R = 5.17 \Omega} \quad (1m)$$

$$X_L = 25.3 \Omega$$

$$X_L = 2\pi f L = 2\pi \times 50 \times L = 25.3$$

$$\therefore \boxed{L = 0.08 \text{ H}} \quad (1m)$$

Q.4 (i) circuit dia. with correct labels - 3m
characteristics with correct labels - 2m

(ii) Diagram - 2M
Explanation - 3M

Q.5 i) Zener diode regulator dia - 1M
Reverse bias characteristics - 1M
Explanation - 3M

ii) Each question 1M